

In one line: A **value type** (struct, enum, tuple) is **copied** on assignment / argument passing — every variable owns an independent value. A **reference type** (class, actor, closure) is **shared** — assignment copies the *pointer*, all variables see one object with an *identity* (===). Default to struct; reach for class when you need identity.

How it works

- **Equality vs identity:** == = same *value* (Equatable, auto-synthesised for structs/enums whose members conform). === = same *object*; classes only, never synthesised.
- **Let differs:** let s = aStruct freezes the *whole value* (no property can change). let c = aClass freezes only the *reference* — c.x = 1 still compiles.
- **mutating:** a struct/enum method *cannot modify self by default* (self is an immutable copy). Mark it mutating → self is passed as inout; it may assign properties or self = ... (enum state change). Callable **only on a var** — on a let it is a compile error. Classes never need it (they mutate through the reference). nonmutating set = storage lives elsewhere (how @State works).
- **inout** = copy-in / copy-out (write-back on return), not a C pointer. Same var passed twice → exclusivity violation.
- **Copy-on-write (COW):** Array, Dictionary, Set, String are structs that wrap a private **class** buffer. b = a copies the header and bumps the buffer's retain count — O(1), shared. The first *mutation* asks isKnownUniquelyReferenced(&buffer): *false* → copy the buffer (O(n)) then write; *true* → write in place. Keep one owner while mutating in a loop, or every append re-copies.
- **Your own struct gets NO COW for free.** Memberwise copy is eager; a stored class property is copied as a *pointer* (shallow) — both copies share it. Want value semantics over a heap object? Write COW yourself (below).
- **Stack vs heap is an optimiser detail**, not the definition: a struct captured by an escaping closure or boxed in an existential lives on the heap; a class may be stack-promoted. Argue *copy vs share*.

Choosing

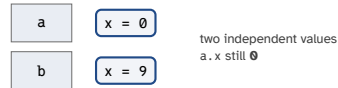
struct (default)	class — only when you need
models, DTOs, geometry, state	identity ("the same" object)
no identity, thread-safe copies	inheritance
Sendable almost for free	ObjC interop / NSObject
SwiftUI state diffs by value	deinit (cleanup)
protocols for polymorphism	shared mutable state, resources

Example — COW by hand

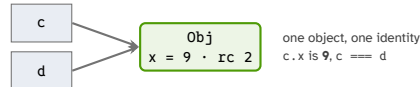
```
final class Box<T> { var v: T; init(_ v: T) { self.v = v } }
struct Pixels {
    private var box: Box<UInt8> // value semantics
    init(_ b: [UInt8]) { box = Box(b) } // shared heap buffer
    var bytes: [UInt8] { box.v } // reads: shared, free
    mutating func set(_ i: Int, _ x: UInt8) {
        if !isKnownUniquelyReferenced(&box) {
            box = Box(box.v) // shared -> copy first
        }
        box.v[i] = x // unique -> in place
    }
}
let p = Pixels([0, 0]); // p.set(0, 9) error: p is a let
```

Copy, share, copy-on-write

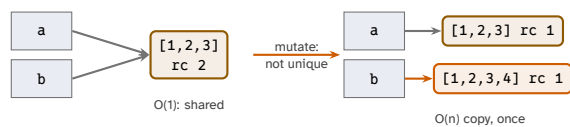
struct: var b = a; b.x = 9



class: let d = c; d.x = 9



Array COW: var b = a, then b.append(4)



struct holding a class — shallow copy



Traps

- ! Calling a mutating method on a let struct — compile error; the same call on a let class instance compiles.
- ! **“Structs are COW.”** Only the stdlib collections (class storage + isKnownUniquelyReferenced); *your own structs do not get it.*
- ! Picking a class for “shared state” alone — the four real reasons are **identity, inheritance, ObjC interop, deinit**; otherwise struct.
- ! “Structs live on the stack” — implementation detail, not semantics.
- ! let arr = [S()]; arr[0].x = 1 — compile error; let freezes the elements too.
- ! Closures are *reference* types: two copies share captured vars.
- ! isKnownUniquelyReferenced takes inout (&box), native Swift classes only; call it inside the mutating method.

What let / var lets you change

	struct (value)	class (reference)
let x	nothing — no property, no rebound	its var properties; no rebound
var x	any property, mutating methods	properties + rebound x
passed to a func	a copy (inout to write back)	the same object

Value: struct enum tuples, Int String Array Dictionary Set Optional. Reference: class actor closures, every NSObject (NSString, NSMutableArray...).

Remember

Struct = photocopy, class = shared online document, COW = photocopy made lazily at the first pen stroke. I-I-O-D — Identity, Inheritance, ObjC, Deinit — the four reasons for a class.